MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG4DSESTA200– Data Analysis Using JAMOVI

Duration: 2 Hours **Maximum: 70 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions
Answer any ten questions. Each question carries **3 marks**.

1. Differentiate between variable types in JAMOVI. [K][1]
2. Briefly explain the Rj Editor module. [U][1]
3. Explain any three functions in analysis tab. [U][1]
4. What is syntax mode in JAMOVI and give its importance. [R][1]
5. Enter the following data and find Pearson's correlation coefficient. Draw a scatter plot and explain strength and direction of the relationship. [A][2]

Study Hours	Exam Score
2	55
4	70
6	82
8	95

6. For the following data fit a simple linear regression model and identify the relationship between age and height. [An][2]

Age	Height
12	75
14	78
16	81
18	84

7. Define log linear regression and give its significance. [U][2]

8. For the data given perform log linear regression and check whether the time of admission and admission type are dependent or independent. [An][2]

Time	Admission Type	Count
Day	Emergency	45
Day	Scheduled	120
Night	Emergency	85
Night	Scheduled	15

9. Define binomial test. [K][4]

10. Exam score of 10 students are given, perform one sample t-test and check whether average score significantly differs from 75. [A][4]

78,82,74,88,79,81,76,85,80,77

11. Define Reliability analysis. [K][5]

12. Why assumption checks in principal component analysis and List any one of them. [U][5]

Part B

Short Essay questions

Answer any four questions. Each question carries **7 marks.**

13. A dataset is given, find the variable names, variable types, data types and add levels if required for all the variables. [An][1]

ID	Group Code	Satisfaction Rank	Score	Active Member
101	1	3	88.5	Yes
102	2	1	72.0	No
103	1	2	91.2	Yes
104	2	2	65.4	Yes
105	3	1	80.0	No

14. A pharmaceutical company has developed a new medication intended to reduce migraine frequency. In a pilot trial, 50 patients were given the medication, and 35 reported meaningful improvement while 15 reported no improvement. The company considers the medication effective only if the improvement rate exceeds the established placebo response rate of 60%. Using a binomial test in Jamovi, determine whether the proportion of patients showing improvement is significantly greater than 60%. [An][4]

15. For the following dataset [An][1]

ID	Treatment type	Pain level	Recovery time	Age	Heart Rate
001	1	2	12	45	72
002	1	3	15	52	75
003	2	1	8	38	68
004	2	2	10	41	70
005	1	3	14	60	80
006	2	1	7	35	65
007	1	2	13	48	74
008	2	1	9	40	69

(a) Apply a filter to only analyze patients aged above 40 and have a heart rate greater than 72. Obtain the new data.

(b) Create a new variable named Heart Rate Target that calculates the target heart rate for moderate exercise which is defined as 70% of existing BPM value.

16. Explain advantages and disadvantages of JAMOVI. [U] [1]
17. For the given data.

GENDER	SLEEP HOURS	CUPS OF COFFEE	PRODUCTIVITY
Female	8	1	85
Male	6	3	70
Female	7	2	78
Male	5	4	62
Female	9	0	92
Male	4	5	55
Female	7	2	80
Male	6	3	72
Female	8	1	88
Male	5	4	65

(a) Explain the variable types and data types in the given dataset.
(b) Using descriptive measures and box plot perform a comparative analysis of productivity scores between the female and male groups.
(c) Interpret the distribution and skewness using histogram and density plot. [An][1]

18. Open HairEyeColor dataset from JAMOVI library
    (a) Explain whether the data is suitable for PCA.
    (b) Determine number of principal components using eigenvalues, scree plot and interpret the results.
    (c) Describe the importance and advantages of PCA. [U][5]

Part C

Essay questions

Answer any one question. Each question carries **12 marks**.

19. (a) Explain in detail one sample test. [A][4]

(b) A researcher wants to know whether there is a significant difference in average daily study hours between male and female undergraduate students.

Data (Study hours per day)	
Male students	3.5, 3.8, 4.0, 3.6, 3.9, 4.1, 3.7
Female students	4.0, 4.2, 4.1, 4.3, 4.0, 4.4, 4.1

i) State the null and alternative hypotheses.
ii) Perform a **two-sample t-test in Jamovi** at **$\alpha = 0.05$**.
iii) Report the **confidence interval** and **p-value**.
iv) Interpret the result in the context of the problem

20. Examine suitability of data for factor analysis. [U][3]

a) Perform exploratory factor analysis using JAMOVI
b) Explain factor loadings and communalities
c) Interpret the results and give conclusion.